

BatchETL Pipeline Architecture

End-To-End Data Engineering Pipeline

Apache Airflow | Pandas | PostgreSQL | Streamlit

LEGEND

- Orchestration
- Processing
- Storage
- Visualization

DOCKER CONTAINER ENVIRONMENT

ORCHESTRATION LAYER (Air Flow)

Dags/eti_pipelines.py

Schedule: @daily | Retries :1 | Owner: Arkan Tsabit

Triggers DAG

PROCESSING LAYER (Python + Pandas)

EXTRACT

scripts/extract.py

TRANSFORM

scripts/transform.py

LOAD

scripts/load.py

Load Data

STORAGE LAYER

EXTRACT

scripts/extract.py

Staging

scripts/extract.py

PostgreSQL 15

Data Warehouse

Query Data

VISUALIZATION LAYER (Streamlit)

Streamlit Dashboard

5KPIs | 4 Charts | 3 Filters | <200ms

Notes :

1. Orchestration Layer: Airflow schedules and monitors the ETL pipeline
2. Processing Layer: Python scripts with Pandas perform data transformations
3. Storage Layer: PostgreSQL stores the final fact_trips table
4. Visualization Layer: Streamlit provides real-time analytics dashboard
5. All services run inside Docker containers for consistency and portability